ANCHOR UNIVERSITY, LAGOS
FACULTY OF SCIENCES
DEPARTMENT OF COMPUTING

2025/2026 FIRST SEMESTER UNIVERSITY EXAMINATION
COURSE TITLE: OPERATING SYSTEMS COURSE CODE: CSC 311 UNIT : 3
TIME : 2 HRS 30 MINS ANSWER ONLY **FOUR** QUESTIONS

Question one

(a) What is an operating system? (3 marks)

(b) Write short note on components of operating system (6 marks)

(c) Mention and discuss functions of operating system (6 marks)

(d) Explain the structures of operating system (5 marks)

Question two

(a) What is a process? (2 marks)

(b) What are the differences between process and program? (6 marks)

(c) Explain if possible with diagram process control block (6 marks)

(d) Write short note on (i) context switch (ii) scheduler (iii) dispatcher (6 marks)

Question three

Write a program in c that would accept burst times with their respective processes indexes using Shortest Job First CPU Scheduling. The program would sort burst times with their processes indexes and calculate waiting time, turn around time, average turn around time and average waiting time as being done in the class (20 marks)

Question four

Write a program in c that would accept both arrival time and burst time using First Come First Serve of randomly arrival times. The program would calculate completion time, turn around time, waiting time, average turn around time and average waiting time as being done in the class (20 marks)

Communication
file management
I/O devices

Question five

Consider the following set of processes with the CPU-burst time and priority scheduling points as shown in table one

Table one: Priority CPU scheduling

Process	Burst Time	Priority
P1	5	1
P2	7	6
P3	2	4
P4	3	5

Use the information in table one to calculate the following:

(a) response time for process P2 and P4 (b) turn around time (c) waiting time

(d) average turn around time (c) average waiting time (4 marks each)

Question six

Consider the arrival time and burst time with time slot of 5 sec for three processes P1, P2 and P4 as illustrated in table two.

Table two: Round Robin CPU scheduling

Process	Arrival Time	Burst time
P1	0	10
P2	1	8
P3	2	7

Use the information in table two to calculate the following:

(a) completion time (b) turn around time (c) waiting time

(d) average turn around time (e) average waiting time (4 marks each)